# Order and Repetition

## Problem 1:

For each of the below problems, first answer the following two questions: "Does the order of the choices matter? (Order:yes or no)" and "Is repetition of choices allowed?" (Repetition: yes or no) Suppose you have a *huge* box of animal crackers containing plenty of each of 10 different animals.

1. How many animal parades containing 6 crackers can you line up?
   - Order:yes or no
   - Repetition: yes or no
2. How many animal parades of 6 crackers can you line up so that the animals appear in alphabetical order?
   - Order:yes or no
   - Repetition: yes or no
3. How many ways could you line up 6 different animals in alphabetical order?
   - Order:yes or no
   - Repetition: yes or no
4. How many ways could you line up 6 different animals if they can come in any order?
   - Order:yes or no
   - Repetition: yes or no
5. How many ways could you give 6 children one animal cracker each?
   - Order:yes or n0
   - Repetition: yes or no
6. How many ways could you give 6 children one animal cracker each so that no two kids get the same animal?
   - Order:yes or no
   - Repetition: yes or no
7. How many ways could you give out 6 giraffes to 10 kids?
   - Order:yes or no
   - Repetition: yes or no

## Problem 2

Create a counting question where the answer is $8 \cdot 3 \cdot 5 \cdot 3$. Create another question where the answer is $8 + 3 + 5 + 3$. Create a third question where the answer is $(8 \cdot 3) + (5 \cdot 3)$.

## Problem 3

Suppose you own a gum ball stand that has 20 varieties of gum balls. Write four different questions, one for each of the four possibilities with order mattering (or not) and repetition allowed (or not), about a customer buying 7 gum balls.